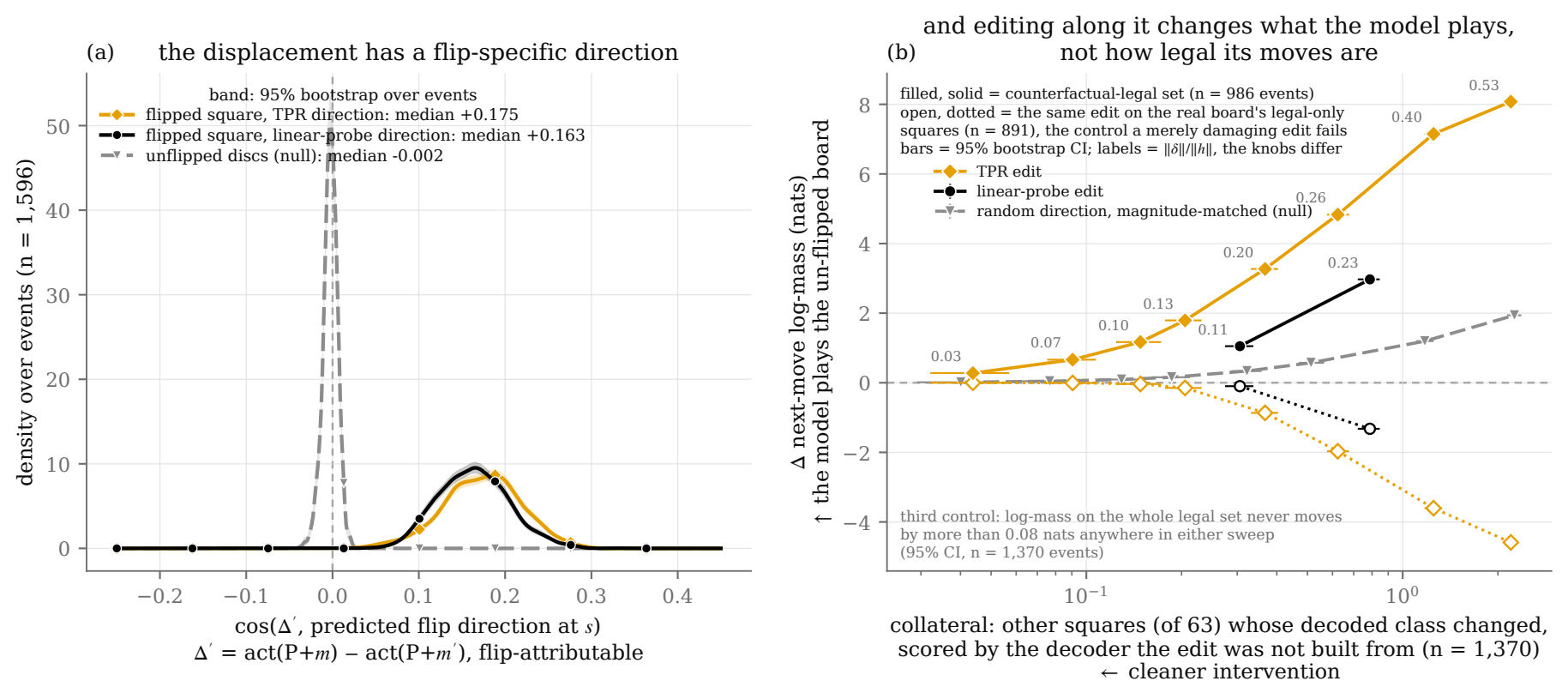
(a) the displacement has a flip-specific direction
band: 95% bootstrap over events
flipped square, TPR direction: median +0.175
flipped square, linear-probe direction: median +0.163
unflipped discs (null): median -0.002
density over events (n = 1,596)
cos(Δ′, predicted flip direction at s)
Δ′ = act(P+m) − act(P+m′), flip-attributable
(b) and editing along it changes what the model plays, not how legal its moves are
filled, solid = counterfactual-legal set (n = 986 events)
open, dotted = the same edit on the real board's legal-only squares (n = 891), the control a merely damaging edit fails
bars = 95% bootstrap CI; labels = ‖δ‖/‖h‖, the knobs differ
TPR edit
linear-probe edit
random direction, magnitude-matched (null)
Δ next-move log-mass (nats)
↑ the model plays the un-flipped board
third control: log-mass on the whole legal set never moves by more than 0.08 nats anywhere in either sweep (95% CI, n = 1,370 events)
collateral: other squares (of 63) whose decoded class changed, scored by the decoder the edit was not built from (n = 1,370)
← cleaner intervention